Structural Induction on Trees

Lecture Notes

Course: Computer-Assisted Theorem Proving (7CCSACTP)

1. Definitions & Recalled Concepts

Functions on Lists & Trees:

- $\texttt{List-set} :: \texttt{'a list} \Rightarrow \texttt{'a set}$

$$\texttt{List-set []} = \emptyset$$
$$\texttt{List-set (x \# xs)} = \text{insert } x \; (\texttt{List-set } xs)$$

- $\texttt{Tree-set} :: \texttt{'a tree} \Rightarrow \texttt{'a set}$

$$\texttt{Tree-set Leaf} = \emptyset$$
$$\texttt{Tree-set (Node l x r)} = \text{insert } x \; (\texttt{Tree-set } l \cup \texttt{Tree-set } r)$$

Inorder Traversal:

- $\texttt{inorder} :: \texttt{'a tree} \Rightarrow \texttt{'a list}$

$$\texttt{inorder Leaf} = []$$
$$\texttt{inorder (Node l x r)} = (\texttt{inorder } l) \mathbin{@} [x] \mathbin{@} (\texttt{inorder } r)$$

Theorem to Prove:

$$\texttt{List-set}(\texttt{inorder } t) = \texttt{Tree-set } t \qquad \text{(by induction on } t\text{)}$$

2. Base Case ($t = \text{Leaf}$)

Goal: $\texttt{List-set}(\texttt{inorder Leaf}) = \texttt{Tree-set Leaf}$

Proof Steps:

1. $\texttt{List-set}(\texttt{inorder Leaf}) = \texttt{List-set []}$ *(by subst inorder.simps(1))*
2. $\cdots = \emptyset$ *(by subst List-set.simps(1))*
3. $\cdots = \texttt{Tree-set Leaf}$ *(by Tree-set.simps(1))* ✓

3. Step Case ($t = \text{Node } t_1 \; a \; t_2$)

Induction Hypotheses:

- $\text{I.H.}_1 : \texttt{List-set}(\texttt{inorder } t_1) = \texttt{Tree-set } t_1$
- $\text{I.H.}_2 : \texttt{List-set}(\texttt{inorder } t_2) = \texttt{Tree-set } t_2$

Goal: $\texttt{List-set}(\texttt{inorder}(\texttt{Node } t_1 \; a \; t_2)) = \texttt{Tree-set}(\texttt{Node } t_1 \; a \; t_2)$

Auxiliary Lemma (Lem 1): $\texttt{List-set}(xs \mathbin{@} ys) = \texttt{List-set } xs \cup \texttt{List-set } ys$

Step-by-Step Equational Proof:

1. $\texttt{List-set}(\texttt{inorder}(\texttt{Node } t_1 \; a \; t_2)) = \texttt{List-set}(\texttt{inorder } t_1 \mathbin{@} [a] \mathbin{@} \texttt{inorder } t_2)$ *(inorder.simps(2))*
2. $\cdots = \texttt{List-set}(\texttt{inorder } t_1) \cup \texttt{List-set}([a] \mathbin{@} \texttt{inorder } t_2)$ *(by Lem 1)*
3. $\cdots = \texttt{List-set}(\texttt{inorder } t_1) \cup \texttt{List-set}[a] \cup \texttt{List-set}(\texttt{inorder } t_2)$ *(by Lem 1)*
4. $\cdots = \texttt{Tree-set } t_1 \cup \texttt{List-set}[a] \cup \texttt{List-set}(\texttt{inorder } t_2)$ *(by I.H.₁)*
5. $\cdots = \texttt{Tree-set } t_1 \cup \text{insert } a \; (\texttt{List-set []}) \cup \texttt{List-set}(\texttt{inorder } t_2)$ *(List-set.simps(2))*
6. $\cdots = \texttt{Tree-set } t_1 \cup \text{insert } a \; \emptyset \cup \texttt{List-set}(\texttt{inorder } t_2)$ *(List-set.simps(1))*
7. $\cdots = \texttt{Tree-set } t_1 \cup \{a\} \cup \texttt{Tree-set } t_2$ *(by I.H.₂)*
8. $\cdots = \{a\} \cup (\texttt{Tree-set } t_1 \cup \texttt{Tree-set } t_2)$
9. $\cdots = \texttt{Tree-set}(\texttt{Node } t_1 \; a \; t_2)$ ✓